Research activity

Cooling of a drink — 40 min.

A drink initially at $80\,^\circ\text{C}$ is placed in a room kept at $20\,^\circ\text{C}$. For $t \geq 0$, let $T(t)$ be its temperature in degrees Celsius, where t is the elapsed time in minutes. We accept the model $T'(t) = -0.1(T(t) - 20)$ and $T(0) = 80$.

1. Solve this initial value problem and give the expression of $T(t)$.
2. Study the direction of variation of T and find its limit at $+\infty$. Interpret the results.
3. Find the first instant at which the temperature is $40\,^\circ\text{C}$. Give the exact value, then an approximate value in minutes.
4. Apply two steps of Euler's method with a step of 1 minute. Compare the approximation obtained after two minutes with the exact value $T(2)$.
5. Give a concrete situation in which the assumption of a constant room temperature would no longer be appropriate.